Overview

Storey 1 · January 2022 · historical data

Zone: All zones | Load type: All loads | Month: January 2022

Data coverage 98.7 %

Electricity used
2,841 kWh
+8.7 % vs December

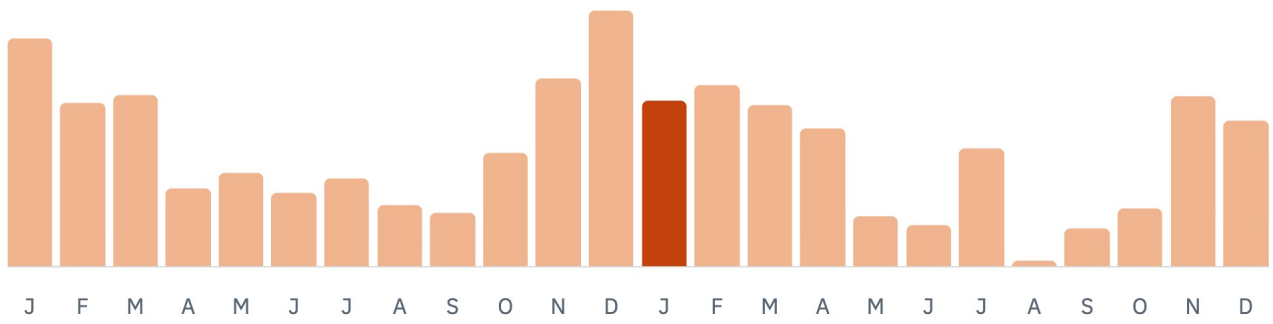
Used while the building is closed
59.5 %
nights, weekends, holidays

Out-of-hours waste
€ 109
765 kWh · 77 kg CO_2 · €168 in December

Alerts this month
64
21 High · 17 Medium · 26 Low

Out-of-hours waste per month (€)

2021–2022 · winter peaks



2021 · 2022 · selected month highlighted

Where is the waste?

NW € 100	NE € 0
SW € 4	SE € 5

Schematic floor plan, not to scale

Top 5 waste sources

NW · Plugs & Sockets	€ 84.1
NW · Lighting	€ 14.7
SE · Plugs & Sockets	€ 4.2
SW · Plugs & Sockets	€ 2.0
SW · Lighting	€ 1.6

Largest alerts

All alerts

NW · Plugs & Sockets — High
Thu 27 Jan 20:00 → Mon 31 Jan 08:00 — € 21.5

NW · Plugs & Sockets — High
Fri 21 Jan 20:00 → Mon 24 Jan 08:00 — € 12.5

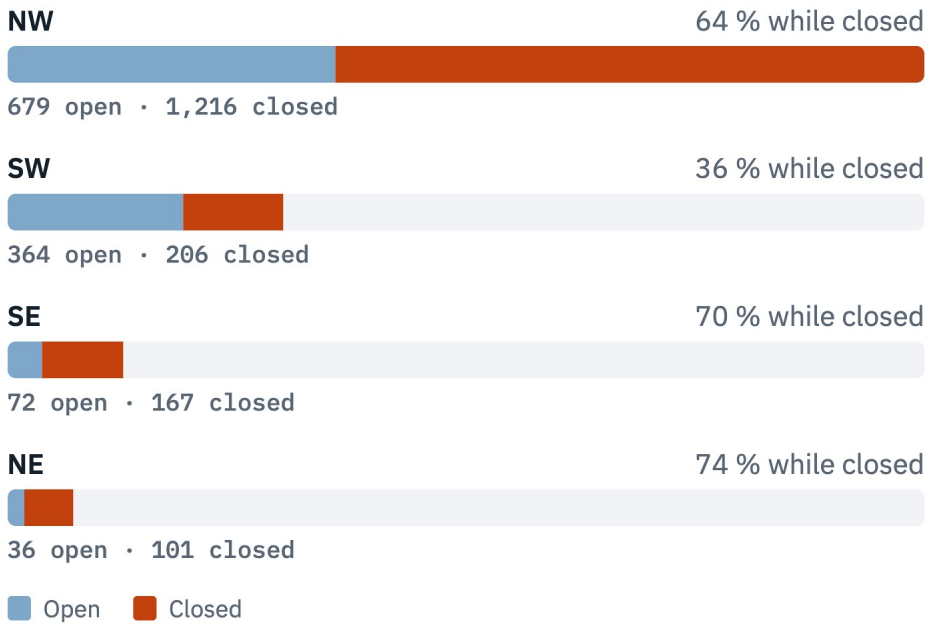
NW · Plugs & Sockets — High
Fri 14 Jan 20:00 → Mon 17 Jan 08:00 — € 10.8

Closed = before 08:00 or from 20:00 on working days, plus weekends, holidays and UPM closures (team calendar). Waste = use above each circuit's standby level (P10) while closed. € at 0.142 €/kWh (Eurostat, all-in) · CO_2 at 0.10 kg/kWh (temporary). Emergency lighting and server rack excluded.

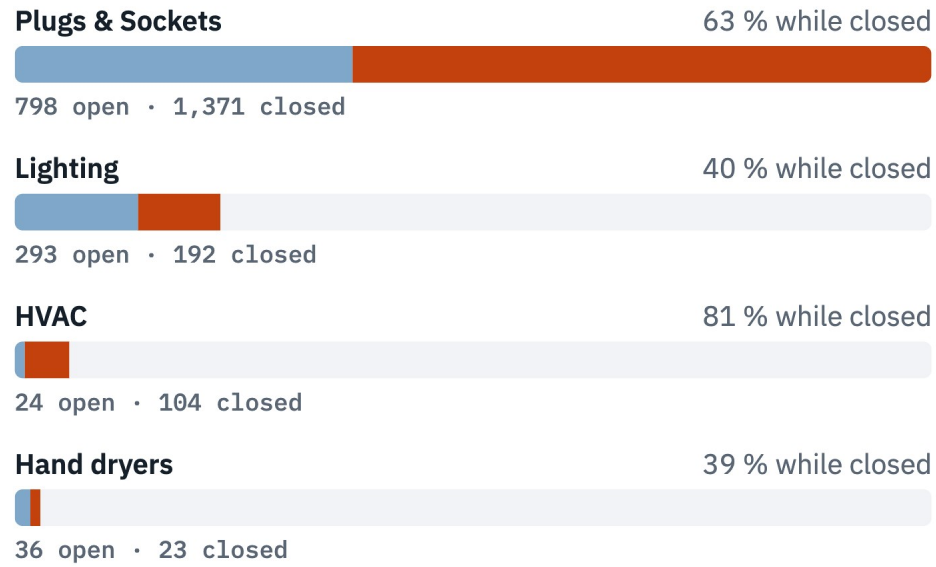
Out-of-hours use

January 2022 · the building was closed 552 of 744 hours (74 %)

Per zone: open vs closed (kWh)



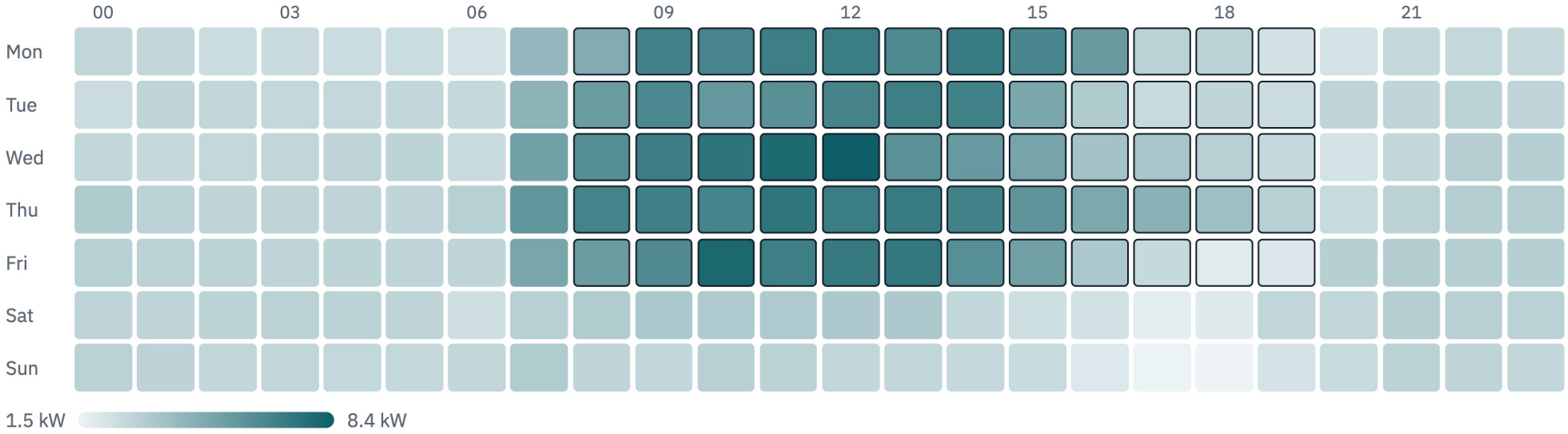
Per load type: open vs closed (kWh)



HVAC runs 81 % of its energy while closed; plugs carry the largest absolute amount (1,371 kWh).

Average power by weekday and hour (kW, whole floor)

Opening hours 08:00–20:00 outlined



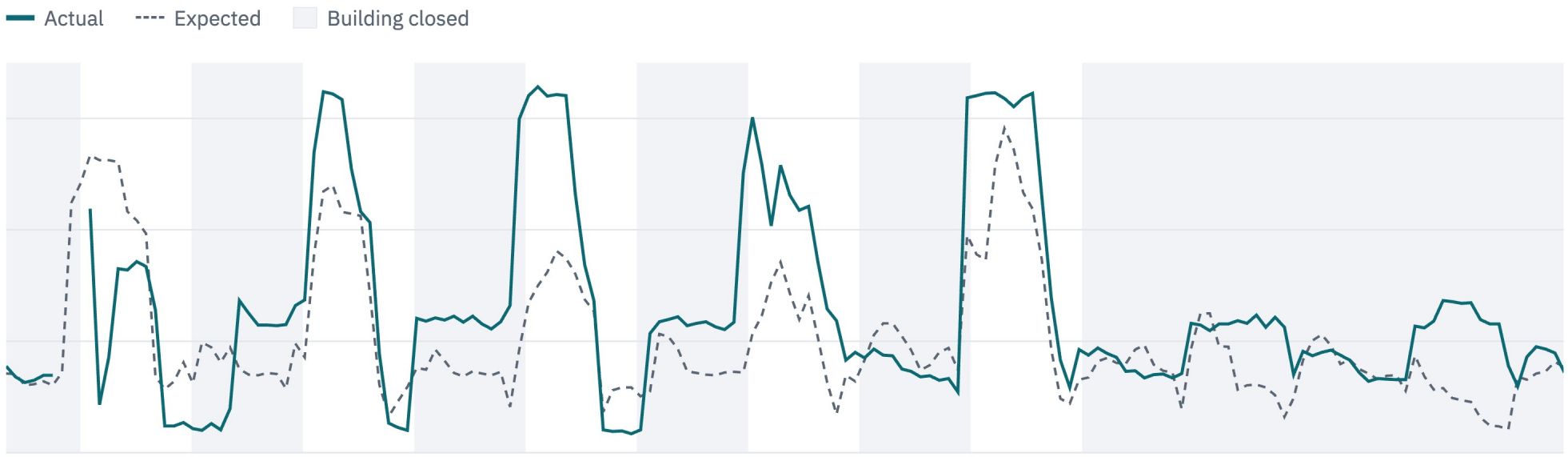
At night and at weekends the floor never drops below about 2.5–3 kW: that standing load is what the waste alerts look at.

Expected vs actual

NW · Plugs & Sockets · week of 10–16 January 2022

Circuit: NW · Plugs & Sockets

Week: 10–16 Jan 2022



Weekday peaks run well above what is expected (up to 6.6 kW), on Friday the load already jumps at 07:00, and at the weekend use stays around 1.8 kW against 1.4 kW expected. Missing hours on Monday morning are shown as a gap, not as zero.

How good is the expectation?

Error as % of average use (CV(RMSE)), NW plugs, validation May–Oct 2022 · lower is better

Same hour last week	81 %
4-week median	68 %
Linear regression (weather + calendar)	51 %
Target (ASHRAE Guideline 14)	≤ 30 %

What this page needs next

- Expected line from the forecast table instead of the simple median
- Prediction band around the expected line
- Alerts marked on the chart, with a link to their detail
- Outdoor temperature as a second, optional line

Expected (sketch) = median of the same weekday and hour over the previous 8 weeks. Model figures from the team's first model notebook (validation period, work in progress).

Alerts

January 2022 · 64 out-of-hours waste alerts · sorted by severity, then €

Out-of-hours waste · 64 | Abnormal use · needs the model | Data errors · for the technician

Severity	Zone · load	When	kWh	€
High	**NW** · Plugs & Sockets	Thu 27 Jan 20:00 → Mon 31 Jan 08:00 84 h	151.6	21.5
High	**NW** · Plugs & Sockets	Fri 21 Jan 20:00 → Mon 24 Jan 08:00 60 h	88.3	12.5
High	**NW** · Plugs & Sockets	Fri 14 Jan 20:00 → Mon 17 Jan 08:00 60 h	75.8	10.8
High	**NW** · Plugs & Sockets	Fri 07 Jan 20:00 → Mon 10 Jan 06:00 58 h	58.9	8.4
High	**NW** · Lighting	Thu 27 Jan 20:00 → Mon 31 Jan 08:00 84 h	30.5	4.3
High	**NW** · Lighting	Fri 21 Jan 20:00 → Mon 24 Jan 08:00 60 h	26.9	3.8

Showing 6 of 64

NW · Plugs & Sockets

High

Thu 27 Jan 20:00 → Mon 31 Jan 08:00 · 84 hours

Excess	Cost	CO_2
152 kWh	€ 21.5	15 kg

Why it was flagged

Average 2.4 kW while closed, about 1.8 kW above this circuit's standby level. A long weekend: Friday 28 January was a UPM non-teaching day (Santo Tomás de Aquino). Outside it was cold: 5.7 °C on average, down to −1.2 °C. Likely heaters left on.

Chart: use 24 h before → 24 h after, standby level as a line

Handled | Under investigation | False alarm | Data error

Severity: High ≥ €2 per alert (or Medium lasting ≥ 24 h), Medium €0.50–2, Low below €0.50. Alerts in this sketch come from a prototype detector run on the team's clean dataset; the real detector (KAN-22) will replace them.

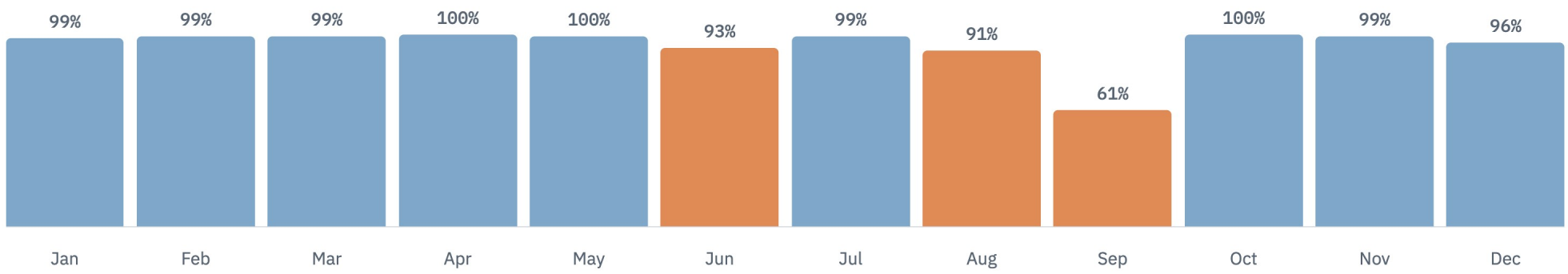
Data quality

Can I trust these numbers? Based on the team's data reliability report.

Reliable	Use with caution	Unreliable	Excluded
18	**6**	**4**	**2**
of 30 columns	e.g. SE HVAC, NE plugs	e.g. NW emergency lighting	SE projector, SW rack

Share of hours with data, 2022 (all circuits)

Below 95 % in orange



September 2022 drops to 61 % because every series is missing from 29 August to 12 September. Figures for such months carry a warning on every page.

Largest gaps

When	What is missing	Hours
Mar 2023 – Aug 2025	All columns, no rows	**21,642**
From 3 Dec 2021	SW server rack, until the end	**13,792**
12 May – 18 Jun 2021	NW emergency lighting and NW HVAC	**887**
23 Dec 2022 – 19 Jan 2023	NW emergency lighting and NW HVAC	**553**